HANZALAH WAHEED

hanzalah.w@gmail.com +971 567975208 GitHub LinkedIn hanzalahwaheed.com Dubai, UAE

SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Web: HTML, CSS, Next.js, Node.js, FastAPI, Nginx
Databases: PostgreSQL, Drizzle ORM, Redis, MongoDB
AI: LLMs, RAG, guardrails, embeddings, pgvector
Cloud: AWS, Vercel, Cloudflare
Tools: Docker, Git, Linux, Claude Code

EXPERIENCE

Founding Software Engineer, *StockInsights.ai* Aug 2024 - Present

- Engineered the platform's **webhook delivery system** end to end (Python, SQS, PostgreSQL, Next.js), implementing **HMAC-SHA256** request signing with rotatable per-subscriber secrets, **at-least-once delivery** with full-jitter retries, crash recovery, SSRF protection, concurrent batched dispatch, and self-service subscription management.
- Led the **MongoDB → PostgreSQL re-platform** across the web application and data pipeline, migrating **25+ collections** through a zero-downtime rollout with backfill and verification tooling, eliminating MongoDB Atlas and Elasticsearch while reducing infrastructure costs by **USD 5,000+/year**.
- Implemented a **RAG pipeline** for financial filings using **OpenAI embeddings, pgvector, and GPT models**, automatically summarizing **25,000+ filings per month** and reducing document analysis from minutes to seconds.
- Architected a **fault-tolerant AI chat platform** (Vercel AI SDK, PostgreSQL) supporting **50,000+ monthly users**, with server-side SSE draining that preserves streamed responses across browser disconnects and reloads.
- Reduced annual infrastructure costs by **approximately USD 4,800** by identifying AI crawler traffic, deploying targeted robots.txt and WAF protections, and optimizing logging.
- Led the **frontend architecture** for StockInsights, leading core user flows, application performance, and **user experience** across the production web application.

Google Summer of Code Contributor, *Ceylon Computer Science Institute* June 2026 - August 2026

- Drove **DataLoom**, a browser-based data-wrangling platform, from a bare CSV table to a full analysis tool with **12+ user-facing features** over **30 merged PRs**, scoping and designing each before building it.
- Consolidated a **dual-endpoint transform API** into a **single registry-dispatched endpoint** covering 21 operations, enforcing **startup-time registry validation** that eliminated an entire class of runtime routing errors.
- Built a type-aware **data profiling engine** for numeric, categorical, boolean, and datetime columns with automatic skew, zero-inflation, cardinality, temporal, and **Pearson correlation** analysis, served through a **single batched API**.
- Authored **1,040 automated tests (691 pytest, 349 Vitest)** covering authentication, transformation semantics, security boundaries, and API behavior.

PROJECTS

prfrd | *Next.js, Vercel AI SDK, TypeScript, PostgreSQL, Drizzle, OpenAI Codex* Repository | Live Demo

- Developed an adversarial **multi-agent analysis engine**: advocate and examiner agents debate opposing, evidence-backed positions on an engineer's quarterly signals, with an arbiter decision and full debate transcripts persisted to PostgreSQL for auditability.
- Defined the signal pipeline and data model: weekly GitHub and Slack activity normalized into PostgreSQL, synthesized into monthly and quarterly evidence that grounds every agent claim.
- Delivered role-based dashboards: manager views with team KPI rollups, risk flags (after-hours load, blockers), and engineer drill-downs backed by the persisted agent outputs.

ProductOwl | *Next.js, TypeScript, BrightData, Cheerio, MongoDB* Repository | Live Demo

- Built a web scraping system using **BrightData** and Cheerio to extract structured product data from Amazon.
- Automated scheduled **price monitoring** with cron jobs, detecting pricing changes in near real time.
- Implemented an automated **email alert workflow** notifying users when tracked prices drop below their targets.

EXTRA-CURRICULAR

- **Open Source:** Sustained contributor across the Python and TypeScript ecosystems: end-to-end feature ownership on DataLoom (data profiling, multi-format ingestion, authentication, workspace UI), feature development and bug fixes on DocsGPT, CircuitVerse and Zulip, and code review for preCICE. Full log: github.com/hanzalahwaheed/OpenSource-Contributions.
- **Achievements:** Won **Best Project Presentation** at the college Open Day for GIST-3DR, a generative implant-segmentation system with integrated 3D and AR visualizations for dental implant planning. Repository

EDUCATION

Dayananda Sagar College of Engineering - Visvesvaraya Technological University Bangalore, India
Bachelor of Engineering in Computer Science and Design *CGPA 9.0/10.0*